QP code: 26900103

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCHE301 -PHYSICAL CHEMISTRY 2

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

INSTRUCTIONS TO THE EXAMINERS

- This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct, may be given due credit.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.
- ***If the answer is complete full credit may be given even if the required number of valuation points could not be identified***

Part A

Multiple Choice Questions

Answer all questions, each question carries 1 mark

1. X-ray diffraction by crystals is mainly used to determine:

c). Crystal structure [U] / [CO1]

2. Silicon doped with arsenic produces a:

b). n-type semiconductor [K] / [CO1]

3. In the smectic phase, the molecules are:

b). Arranged in layers with their long axes approximately parallel [K] / [CO1]

4. In the photosensitized isomerization of 2-butene, the sensitizer

a). Absorbs light and transfers excitation energy to 2-butene [U] / [CO2]

5. In a Jablonski diagram, the lowest electronic singlet state is represented as:

a). S_0 [U] / [CO2]

6. For an isochoric process, the internal energy change is equal to

c). q [U] / [CO3]

7. When work is done on a system under adiabatic condition, its internal energy

b). Increases [An] / [CO3]

8. The area enclosed by a Carnot cycle on a P-V diagram represents

c). Net work done [U] / [CO4]

9. When a real gas shows a cooling effect during expansion through a porous plug, the Joule Thomson coefficient is

a). Positive [U] / [CO3]

10. A substance having disorder or degeneracy even at 0 K may possess

Out of syllabus, give full credit to all [U] / [CO4] **[1x10 = 10]**

Part B

Short Answer Questions

Answer any 4 questions, each question carries 5 marks

11. Briefly explain the structure of zinc blende. [U] / [CO1]

Zinc Blende has a cubic close packed structure (FCC lattice). Sulfide ions sit at the corners and the centers of each face of the cubic unit cell. Zinc ions occupy half of the tetrahedral voids. Both Zn^{2+} and S^{2-} have a 4:4 coordination number. There are 4 formula units per unit cell. ***(Give full credit for any five relevant points)***

12. Explain the Schottky defect. [K] / [CO1]

A Schottky defect is a type of point defect in an ionic crystal in which an equal number of cations and anions are missing from their normal lattice sites.

- Equal numbers of positive and negative ions are absent from the crystal lattice.
- Electrical neutrality is maintained because the number of missing cations and anions is equal.
- The density of the crystal decreases because some ions are missing from the lattice.
- The stoichiometry of the crystal remains unchanged. Examples: NaCl, KCl, KBr and CsCl.

(proper explanation can be given full credit, illustration with diagram can also be considered for giving credit)

13. Calculate the energy per mole of light having wavelengths of (a) 85 nm and (b) 300 nm.

$$E = \frac{N_A hc}{\lambda}$$ [U] / [CO2]

a) Given: Planck's constant, h = 6.626 x10^{-34} Js, Speed of light, c = 3.0 x10^{8}m s^{-1}, Avogadro constant, N_A = 6.022 x $10^{23}$$mol^{-1}$.

$$\lambda = 85\ nm = 85 \times 10^{-9}\ \text{m}$$

Substituting the values $E = 1.410 \times 10^{6}\ \text{J mol}^{-1}$ or $E = 1410\ k\text{J mol}^{-1}$

b) $\lambda = 300\ nm = 300 \times 10^{-9}$ m

$E = 3.989 \times 10^{5}\ \text{J mol}^{-1}$ or $E = 398.9\ \text{kJ mol}^{-1}$

(Equation and correct answer with proper units can be given full credit, equation alone -2 marks)

14. What are primary and secondary processes in photochemical reactions? [U] / [CO2]

1. Primary Process

The primary process is the initial step in a photochemical reaction in which a molecule absorbs light energy and becomes electronically excited. The excited molecule may undergo dissociation, ionization, isomerization, or other photochemical changes.

The secondary process consists of the subsequent chemical reactions involving the products formed during the primary process. These reactions may occur with or without further absorption of light. *(Give any suitable example-general reaction can also be given credit)* ***(Any reasonable explanation can be given full credits)***

15. State and explain the different statements of the second law of thermodynamics. [U] / [CO3]

Kelvin–Planck Statement: It is impossible to construct a cyclic heat engine that converts all the heat absorbed from a single heat reservoir completely into work.

Clausius Statement: It is impossible to construct a cyclic device whose sole effect is to transfer heat from a colder body to a hotter body without external work.

Entropy Statement: The entropy of an isolated system never decreases in a spontaneous process.

(Any two correct statements with short explanation can be given full credit, statement alone-3marks)

16. Derive the expression for entropy as a function of volume and temperature. [A] / [CO4]

From second law: $dS = \frac{dq_{rev}}{T}$

From first law: $dq_{rev} = dU + PdV \qquad dU = C_v dT \qquad \Rightarrow dq_{rev} = C_v dT + PdV$

For 1 mole of gas: $dq_{rev} = C_v dT + \frac{RT}{V} dV \qquad \frac{dq_{rev}}{T} = \frac{C_v}{T} dT + \frac{R}{V} dV$

For Finite Change: $\int_{S_1}^{S_2} dS = C_v \int_{T_1}^{T_2} \frac{dT}{T} + R \int_{V_1}^{V_2} \frac{dV}{V} \qquad \Delta S = C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1}$

Or in log form: $\Delta S = 2.303\, C_v \log \frac{T_2}{T_1} + 2.303\, R \log \frac{V_2}{V_1}$

For *n* moles of gas: $\Delta S = 2.303\, nC_v \log \frac{T_2}{T_1} + 2.303\, nR \log \frac{V_2}{V_1}$

(Proper derivation will fetch full credit) **[5x4=20]**

Part C

Short Essay

Answer any 2 questions, each question carries 10 marks

17. (a) Explain in detail the law of rational indices. (b) Write a short note on Weiss indices and Miller indices. [U] / [CO1]

(a) Law of Rational Indices: The intercepts made by any crystal face on the crystallographic axes, when expressed in terms of the unit cell parameters, are either infinite or simple rational multiples of those parameters. If a crystal face cuts the three crystallographic axes at the following distances: Along the a-axis: 2a, Along the b-axis: 3b, Along the c-axis: c. The intercepts are therefore:2a:3b:c. ***(4 marks)***

(b) **Weiss indices**: Weiss indices represent a crystal face by expressing its intercepts on the crystallographic axes as multiples of the unit cell dimensions.

Method: Determine the intercepts of the crystal face along the a, b and c axes. Express the intercepts in terms of the corresponding unit cell parameters. Write the intercepts as a ratio. Example: If a crystal face cuts the axes at 2a, 3b and c, its Weiss indices are [2:3:1].

Miller indices: Miller indices are a set of three smallest integers, represented as (hkl), used to specify the orientation of a crystal plane.

Steps for determining Miller indices: Determine the intercepts of the plane along the a, b and c axes in terms of the unit cell parameters. Take the reciprocals of the intercepts. Multiply the reciprocals by a suitable common factor to obtain the smallest whole numbers. Enclose the resulting integers in parentheses as (hkl).

Example: Suppose a crystal face has intercepts 2a, 3b and c. The intercepts are: [2,3,1]. Their Miller indices are (326). ***(6 marks)*** ***(Reasonable explanation with suitable example can be given full credits)***

18. (a) Derive the expression for maximum work obtained during the reversible expansion of an ideal gas from V_1 to V_2 at constant temperature. (b) One mole of an ideal gas expands reversibly and isothermally at 300 K from 5 L to 20 L. Calculate the maximum work done. [K] / [CO3]

(a) Consider a gas enclosed in a cylinder fitted with a frictionless piston. Let the pressure of the gas be P. The work done by the system in increasing the volume by an infinitesimally small amount dV is

$$dw = -P\,dV$$

The total work done during a reversible expansion from initial volume V_1 to final volume V_2 is

$$w_{\text{rev}} = -\int_{V_1}^{V_2} P\,dV$$

For 1 mole of an ideal gas, $P = \frac{RT}{V}$

Since temperature remains constant during an isothermal process,

$$w_{\text{rev}} = -\int_{V_1}^{V_2} \frac{RT}{V}\,dV = -RT\ln\frac{V_2}{V_1}$$

Hence, $\boxed{w_{\text{rev}} = -RT\ln\frac{V_2}{V_1}}$ or $\boxed{w_{\text{rev}} = -2.303\,\text{n}RT\log\frac{V_2}{V_1}}$ ***(6 marks)***

(b) $w_{max} = w_{\text{rev}} = -2.303\,\text{n}RT\log\frac{V_2}{V_1}$

$W_{\text{max}} = -2.303 \times 1 \times 8.314 \times 300\log\frac{20}{5} = -3458\,J = -3.458kJ$ ***(4 marks)***

19. Derive Gibbs-Helmholtz equation. [U] / [CO4]

Gibbs energy with temp at const pressure: $G = H - TS$

$\because H = U + PV$ $\quad G = U + PV - TS$

$dG = dU + PdV + VdP - TdS - SdT$

From 1st law $dq = dU + PdV$ $\quad$ From 2nd law $dq_{rev} = TdS$

Thus $TdS = dU + PdV$ $\quad$ Subst in eqn for dG, $dG = VdP - SdT$

At const pressure $dG = -SdT$ $(dP = 0)$ $\quad \Rightarrow \left(\frac{\partial G}{\partial T}\right)_P = -S$

For state 1: $\left(\frac{\partial G_1}{\partial T}\right)_P = -S_1$ $\quad$ state 2: $\left(\frac{\partial G_2}{\partial T}\right)_P = -S_2$

$\left(\frac{\partial G_2}{\partial T}\right)_P - \left(\frac{\partial G_1}{\partial T}\right)_P = -(S_2 - S_1)$ $\quad \left(\frac{\partial (G_2 - G_1)}{\partial T}\right)_P = -\Delta S$ $\quad \left(\frac{\partial (\Delta G)}{\partial T}\right)_P = -\Delta S$

We know that $\Delta G = \Delta H - T\Delta S$ $\quad \Delta G = \Delta H + T\left(\frac{\partial (\Delta G)}{\partial T}\right)_P$

This is known as **Gibbs-Helmholtz equation** in terms of Gibbs energy and enthalpy change at constant pressure. ***(Derivation with proper steps can be given full credit)***

20. Derive Gibbs-Duhem equation. **[U] / [CO4]**

Gibbs energy (G) is a function of T, P and the number of moles $(n_1, n_2, \ldots)$ of the constituents present.

$G = \phi(T, P, n_1, n_2, \ldots)$

A small change in the Gibbs energy dG (a complete differential) can be expressed as:

$$dG = \left(\frac{\partial G}{\partial T}\right)_{P,n_1,n_2,\ldots} dT + \left(\frac{\partial G}{\partial P}\right)_{T,n_1,n_2,\ldots} dP + \left(\frac{\partial G}{\partial n_1}\right)_{T,P,n_2,n_3,\ldots} dn_1 + \cdots$$

Representing the $(\partial G/\partial n_i)$ terms by μ_i, eqn. (b) may be written as:

$$dG = \left(\frac{\partial G}{\partial T}\right)_{P,n_1,n_2,\ldots} dT + \left(\frac{\partial G}{\partial P}\right)_{T,n_1,n_2,\ldots} dP + \mu_1 dn_1 + \mu_2 dn_2 + \cdots$$

For a system at constant T and P (i.e., when $dT = 0$ and $dP = 0$), eqn. (c) simplifies to:

$$dG = \mu_1 dn_1 + \mu_2 dn_2 + \cdots$$

For a system of definite composition, integration of eqn. (d) gives: $G = \mu_1 n_1 + \mu_2 n_2 + \cdots$

When eqn. (e) is differentiated completely, we get: $dG = \mu_1 dn_1 + n_1 d\mu_1 + \mu_2 dn_2 + n_2 d\mu_2 + \ldots$

But according to eqn. (c), for a system of definite composition at constant T and P:

$$dG = \mu_1 dn_1 + \mu_2 dn_2 + \cdots$$

Comparing, we get: $n_1 d\mu_1 + n_2 d\mu_2 + \cdots = 0$ $\quad \sum n_i \, d\mu_i = 0$

This equation is called Gibbs–Duhem equation. ***(Derivation with proper steps can be given full credit)***

[10x2 = 20]